



PES UNIVERSITY

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Department of Computer Science & Engineering

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UE24MA241B: Linear Algebra and its Applications

Classwork Problems

Geometry of Linear Equations

Problem for Chapter 1.2:

Sketch these three lines and decide if the equations are solvable:

3 by 2 system

$$x + 2y = 2$$

$$x - y = 2$$

$$y = 1.$$

What happens if all right-hand sides are zero? Is there any nonzero choice of right-hand sides that allows the three lines to intersect at the same point?

For the equations $x+y = 4$, $2x-2y = 4$, draw the row picture (two intersecting lines) and the column picture (combination of two columns equal to the column vector $(4,4)$ on the right side)

Describe the intersection of the three planes $u+v+w+z = 6$ and $u+w+z=4$ and $u+w=2$ (all in four-dimensional space). Is it a line or a point or an empty set? What is the intersection if the fourth plane $u = -1$ is included? Find a fourth equation that leaves us with

no solution.

These equations are certain to have the solution $x=y=0$. For which values of a is there a whole line of solutions? $ax + 2y = 0$ $2x + ay = 0$

Explain why the system

$$\begin{array}{rcrcrcrcrcl} u & + & v & + & w & = & 2 \\ u & + & 2v & + & 3w & = & 1 \\ & & v & + & 2w & = & 0 \end{array}$$

is singular by finding a combination of the three equations that adds up to $0 = 1$. What value should replace the last zero on the right side to allow the equations to have solutions—and what is one of the solutions?

The column picture for the previous exercise (singular system) is

$$u \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + v \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + w \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = b.$$

Show that the three columns on the left lie in the same plane by expressing the third column as a combination of the first two. What are all the solutions (u, v, w) if b is the zero vector $(0, 0, 0)$?

Gaussian elimination

CHAPTER 1.3

q1

For which numbers a does elimination break down (a) permanently, and (b) temporarily?

$$\begin{array}{rcrcrcrcl} ax & + & 3y & = & -3 \\ 4x & + & 6y & = & 6. \end{array}$$

Solve for x and y after fixing the second breakdown by a row exchange.

Which number q makes this system singular and which right-hand side t gives it infinitely many solutions? Find the solution that has $z = 1$.

$$\begin{aligned}x + 4y - 2z &= 1 \\x + 7y - 6z &= 6 \\3y + qz &= t.\end{aligned}$$

Apply elimination and back-substitution to solve

$$\begin{aligned}2u + 3v &= 0 \\4u + 5v + w &= 3 \\2u - v - 3w &= 5.\end{aligned}$$

What are the pivots? List the three operations in which a multiple of one row is subtracted from another.

Solve the system and find the pivots when

$$\begin{aligned}2u - v &= 0 \\-u + 2v - w &= 0 \\-v + 2w - z &= 0 \\-w + 2z &= 5.\end{aligned}$$

You may carry the right-hand side as a fifth column (and omit writing u, v, w, z until the solution at the end).

Use elimination to solve

$$\begin{array}{rclcl}u + v + w &= & 6 & & u + v + w = 7 \\u + 2v + 2w &= & 11 & \text{and} & u + 2v + 2w = 10 \\2u + 3v - 4w &= & 3 & & 2u + 3v - 4w = 3.\end{array}$$

For which three numbers a will elimination fail to give three pivots?

$$ax + 2y + 3z = b_1$$

$$ax + ay + 4z = b_2$$

$$ax + ay + az = b_3.$$

Elimination Matrices

Section 1.4: Start from the topic- The matrix Form of one Elimination Step(Pg no 21 of 4th Edition)

Chapter 1.4

Which three matrices E_{21} , E_{31} , E_{32} put A into triangular form U ?

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix} \quad \text{and} \quad E_{32}E_{31}E_{21}A = U.$$

Multiply those E 's to get one matrix M that does elimination: $MA = U$.

This 4 by 4 matrix needs which elimination matrices E_{21} and E_{32} and E_{43} ?

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

Triangular Factors and Row Exchanges

Problem set 1.5:

Apply elimination to produce the factors L and U for

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 4 \\ 1 & 4 & 8 \end{bmatrix}.$$

Which numbers a, b, c lead to row exchanges? Which make the matrix singular?

$$A = \begin{bmatrix} 1 & 2 & 0 \\ a & 8 & 3 \\ 0 & b & 5 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} c & 2 \\ 6 & 4 \end{bmatrix}.$$

Find the $PA = LDU$ factorizations (and check them) for

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}.$$

(Recommended) Compute L and U for the symmetric matrix

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}.$$

Find four conditions on a, b, c, d to get $A = LU$ with four pivots.

INVERSES AND TRANSPOSES

Problem Set 1.6

Use the Gauss-Jordan method to invert

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Under what conditions on their entries are A and B invertible?

$$A = \begin{bmatrix} a & b & c \\ d & e & 0 \\ f & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & e \end{bmatrix}.$$

Chapter 2.1

Q1

Describe the column space and the nullspace of the matrices

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 2 & 3 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Q2

Which of the following descriptions are correct? The solutions x of

$$Ax = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

form

- (a) a plane.
- (b) a line.
- (c) a point.
- (d) a subspace.
- (e) the nullspace of A .
- (f) the column space of A .

Q3

Show that the set of nonsingular 2 by 2 matrices is not a vector space. Show also that the set of *singular* 2 by 2 matrices is not a vector space.

Q4

Describe the column spaces (lines or planes) of these particular matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

Q5

Why isn't $\mathbf{R}^2$ a subspace of $\mathbf{R}^3$?

Q6

Construct a 3 by 3 matrix whose column space contains $(1, 1, 0)$ and $(1, 0, 1)$ but not $(1, 1, 1)$. Construct a 3 by 3 matrix whose column space is only a line.

List of Problems for Unit 2

Problem Set 2.2

Q5. (Hard Copy Question #)

2. Reduce A and B to echelon form, to find their ranks. Which variables are free?

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

Find the special solutions to $Ax = 0$ and $Bx = 0$. Find all solutions.

Q12. (Hard Copy Question #)

8. Under what conditions on b_1 and b_2 (if any) does $Ax = b$ have a solution?

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 7 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

Find two vectors in the nullspace of A , and the complete solution to $Ax = b$.

Q26. (Hard Copy Question #)

19. What are the special solutions to $Rx = 0$ and $R^T y = 0$ for these R ?

$$R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Q29. (Hard Copy Question #)

(Recommended) Execute the six steps following equation (6) to find the column space and nullspace of A and the solution to $Ax = b$:

$$A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}.$$

Q32. (Hard Copy Question #)

31. For every c , find R and the special solutions to $Ax = 0$:

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 1 & c & 2 & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1-c & 2 \\ 0 & 2-c \end{bmatrix}.$$

Q33.

36. Which vectors (b_1, b_2, b_3) are in the column space of A ? Which combinations of the rows of A give zero?

$$(a) \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 3 \\ 0 & 2 & 5 \end{bmatrix} \quad (b) \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}.$$

Q56

48. Reduce to $Ux = c$ (Gaussian elimination) and then $Rx = d$:

$$Ax = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix} = b.$$

Find a particular solution x_p and all nullspace solutions x_n .

Problem Set 2.3

Q3.

5. Decide the dependence or independence of

(a) the vectors $(1, 3, 2)$, $(2, 1, 3)$, and $(3, 2, 1)$.

(b) the vectors $(1, -3, 2)$, $(2, 1, -3)$, and $(-3, 2, 1)$.

Q4.

1. Show that v_1, v_2, v_3 are independent but v_1, v_2, v_3, v_4 are dependent:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_4 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

Solve $c_1v_1 + \cdots + c_4v_4 = 0$ or $Ac = 0$. The v 's go in the columns of A .

Q5

7. If w_1, w_2, w_3 are independent vectors, show that the differences $v_1 = w_2 - w_3$, $v_2 = w_1 - w_3$, and $v_3 = w_1 - w_2$ are *dependent*. Find a combination of the v 's that gives zero.

Q7.

2. Find the largest possible number of independent vectors among

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad v_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad v_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \quad v_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

This number is the ____ of the space spanned by the v 's.

Q9.

10. Find two independent vectors on the plane $x + 2y - 3z - t = 0$ in $\mathbf{R}^4$. Then find three independent vectors. Why not four? This plane is the nullspace of what matrix?

Q10.

8. If w_1, w_2, w_3 are independent vectors, show that the sums $v_1 = w_2 + w_3$, $v_2 = w_1 + w_3$, and $v_3 = w_1 + w_2$ are *independent*. (Write $c_1v_1 + c_2v_2 + c_3v_3 = 0$ in terms of the w 's. Find and solve equations for the c 's.)

Q13.

16. Decide whether or not the following vectors are linearly independent, by solving $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

Decide also if they span $\mathbf{R}^4$, by trying to solve $c_1v_1 + \cdots + c_4v_4 = (0, 0, 0, 1)$.

Q15.

13. Find the dimensions of (a) the column space of A , (b) the column space of U , (c) the row space of A , (d) the row space of U . Which two of the spaces are the same?

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Q25

20. Find a basis for each of these subspaces of $\mathbf{R}^4$:

- (a) All vectors whose components are equal.
- (b) All vectors whose components add to zero.
- (c) All vectors that are perpendicular to $(1, 1, 0, 0)$ and $(1, 0, 1, 1)$.
- (d) The column space (in $\mathbf{R}^2$) and nullspace (in $\mathbf{R}^5$) of $U = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$.

Q38.

39. The cosine space $\mathbf{F}_3$ contains all combinations $y(x) = A\cos x + B\cos 2x + C\cos 3x$. Find a basis for the subspace that has $y(0) = 0$.

Problem Set 2.4

Q9

14. Find a left-inverse and/or a right-inverse (when they exist) for

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad M = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad T = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}.$$

Q11

17. Find a matrix A that has $\mathbf{V}$ as its row space, and a matrix B that has $\mathbf{V}$ as its nullspace, if $\mathbf{V}$ is the subspace spanned by

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}.$$

Q13

18. Find a basis for each of the four subspaces of

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Q28

29. find bases for the four fundamental subspaces:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 9 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Problem Set 2.6

Q2

7. On the space $\mathbf{P}_3$ of cubic polynomials, what matrix represents d^2/dt^2 ? Construct the 4 by 4 matrix from the standard basis $1, t, t^2, t^3$. Find its nullspace and column space. What do they mean in terms of polynomials?

Q6

5. The matrix $A = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ yields a **shearing** transformation, which leaves the y -axis unchanged. Sketch its effect on the x -axis, by indicating what happens to $(1,0)$ and $(2,0)$ and $(-1,0)$ —and how the whole axis is transformed.

Q11

8. From the cubics $\mathbf{P}_3$ to the fourth-degree polynomials $\mathbf{P}_4$, what matrix represents multiplication by $2 + 3t$? The columns of the 5 by 4 matrix A come from applying the transformation to $1, t, t^2, t^3$.

Q17

25. Which of these transformations satisfy $T(v + w) = T(v) + T(w)$, and which satisfy $T(cv) = cT(v)$?

- (a) $T(v) = v/\|v\|$. (b) $T(v) = v_1 + v_2 + v_3$.
 (c) $T(v) = (v_1, 2v_2, 3v_3)$. (d) $T(v) = \text{largest component of } v$.

Q18

15. The space of all 2 by 2 matrices has the four basis “vectors”

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

For the linear transformation of *transposing*, find its matrix A with respect to this basis. Why is $A^2 = I$?

Q22

24. Suppose $T(v) = v$, except that $T(0, v_2) = (0, 0)$. Show that this transformation satisfies $T(cv) = cT(v)$ but not $T(v + w) = T(v) + T(w)$.

Q26

27. The “cyclic” transformation T is defined by $T(v_1, v_2, v_3) = (v_2, v_3, v_1)$. What is $T(T(T(v)))$? What is $T^{100}(v)$?

Q30

28. Find the *range* and *kernel* (those are new words for the column space and nullspace) of T .

- (a) $T(v_1, v_2) = (v_2, v_1)$. (b) $T(v_1, v_2, v_3) = (v_1, v_2)$.
 (c) $T(v_1, v_2) = (0, 0)$. (d) $T(v_1, v_2) = (v_1, v_1)$.

Q36

- 36.** (a) What matrix transforms $(1, 0)$ into $(2, 5)$ and transforms $(0, 1)$ to $(1, 3)$?
 (b) What matrix transforms $(2, 5)$ to $(1, 0)$ and $(1, 3)$ to $(0, 1)$?

2.6 Linear Transformations

- (c) Why does no matrix transform $(2, 6)$ to $(1, 0)$ and $(1, 3)$ to $(0, 1)$?

Q41

- 46.** Show that the product ST of two reflections is a rotation. Multiply these reflection matrices to find the rotation angle:

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}.$$

Q49

- 50.** (Recommended) Suppose all vectors x in the unit square $0 \leq x_1 \leq 1$, $0 \leq x_2 \leq 1$ are transformed to Ax (A is 2 by 2).
- (a) What is the shape of the transformed region (all Ax)?
 - (b) For which matrices A is that region a square?
 - (c) For which A is it a line?
 - (d) For which A is the new area still 1?



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Department of Computer Science and Engineering

UE24MA241B - Linear Algebra and its Applications

Unit3 Classwork problem solution

1. Find the length of $a = (2, -2, 1)$, and write two independent vectors that are perpendicular to a . Find the length of a using the formula for vector length.

$$\|a\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Length of $a = 3$

write two independent vectors that are perpendicular to a

1. Write a as a row matrix. $A = [2 \ -2 \ 1]$
 2. Perpendicular vectors lie in the nullspace of A .
 3. Find Nullspace of A
- Final perpendicular vectors
- $(1, 1, 0)$
- $(-1/2, 0, 1)$

-
2. Show that the vectors $v_1 = (1, 2, -1)$, $v_2 = (2, -1, 0)$ are orthogonal.

Check by using formula $v_1^T v_2 = 0$

3. Find a vector in $\mathbb{R}^3$ that is orthogonal to both $(1, 1, 0)$ and $(1, 0, 1)$

Step 1: Let the required vector be $x = (x_1, x_2, x_3)$.

Step 2: Since the vector is orthogonal to $(1, 1, 0)$,

$$(x_1, x_2, x_3) \cdot (1, 1, 0) = 0$$

Step 3: Since the vector is orthogonal to $(1, 0, 1)$,

$$(x_1, x_2, x_3) \cdot (1, 0, 1) = 0$$

Step 4: Solve the equations.

Final vector orthogonal to both vectors:

$$(-1, 1, 1)$$

4. Find a vector x orthogonal to the row space of A , a vector y orthogonal to the column space of A , a vector z orthogonal to the nullspace of A .

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

Vector x orthogonal to the row space

Step 1: A vector orthogonal to the row space lies in the nullspace of A .

Step 2: Solve $Ax = 0$.

Step 3: Row reduce A .

Step 4: Write equations from the reduced matrix.

Step 5: Express pivot variables in terms of the free variable.

Step 6: Choose a value for the free variable to obtain x .

Final: $x = (-2, 1, 0)$

Vector y orthogonal to the column space

Step 1: A vector orthogonal to the column space lies in the nullspace of A^T .

Step 2: Perform Row transformation on A

Step 3: Express the variables in terms of multipliers of row transformation

Final: $y = (-1, -1, 1)$

Vector z orthogonal to the nullspace

Step 1: The orthogonal complement of the nullspace is the row space of A .

Step 2: Take any nonzero row of A .

Final: $z = (1, 2, 1)$

5. Find a basis for the orthogonal complement of the row space of

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix}$$

Split the vector $x = (3, 3, 3)$

Into $x = x_r + x_n$

where x_r is in the row space of A and x_n is in the nullspace of A .

Part 1: Basis for the orthogonal complement of the row space

Step 1

The orthogonal complement of the row space of A is the nullspace of A .

Step 2

Solve

$$Ax = 0$$

Basis for the orthogonal complement

$$\left\{ \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \right\}$$

Part 2: Split $x = (3,3,3)$

Let

$$x_n = c(-2, -2, 1)$$

and

$$x_r = x - x_n$$

Step 1: Write $x_r = (3,3,3) - c(-2, -2, 1)$

Step 2: Since x_r is in the row space $x_r = ar_1 + br_2$ (where r_1 and r_2 are basis vectors of row space)

Step 3: Equate step 1 and step 2, we get $c=-1$

Step 4: Compute components $x_n = -1(-2, -2, 1) = (2, 2, -1)$ $x_r = x - x_n$

Basis for orthogonal complement of row space

$$\left\{ \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \right\}$$

$$x_r = (1, 1, 4)$$

$$x_n = (2, 2, -1)$$

6. Let P be the plane in $\mathbb{R}^3$ with equation

$$x + 2y - z = 0$$

Find a vector perpendicular to P .

Find a matrix whose nullspace is the plane P .

Find a matrix whose row space is the plane P .

Step 1: Write the equation in matrix form

$$\begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Basis vectors of the plane is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

i) vector perpendicular to P .

The coefficients of the plane equation give the normal vector.

ii) Basis vectors of the plane p is Nullspace of $\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$. Hence the plane is the nullspace of $\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$

iii) Basis vectors of the plane p is Row space of $\begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. Hence the plane is the row space of $\begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

7. Find the projection of $b=(1,2,7)$ onto the column space of A spanned by $(1,1,-2)$ and $(1,-1,4)$ split b into $p+q$ with p in $C(A)$ and q in Left Null Space of A .

Step 1: Find the basis of the left nullspace of A

Basis of left nullspace

$$y = (-1, 3, 1)$$

Step 2: Find projection of b on the left nullspace basis

Projection formula

$$\text{proj}_y(b) = \frac{b \cdot y}{y \cdot y} y$$

Left nullspace component of b

$$q = \left(-\frac{12}{11}, \frac{36}{11}, \frac{12}{11} \right)$$

Step 3: Column space component of b

$$p = b - q$$

$$p = \left(\frac{23}{11}, -\frac{14}{11}, \frac{65}{11} \right)$$

8. Find the matrix that projects every point in the plane onto the line $x+2y=0$

Step 1: Find a basis vector for the line $x + 2y = 0$

Basis vector for the line:

$$v = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Step 2: Find Projection matrix onto a line spanned by v using

$$P = \frac{vv^T}{v^T v}$$

$$P = \begin{bmatrix} \frac{4}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}$$

This matrix projects every point in the plane onto the line $x + 2y = 0$.

9. What multiple of $a=(1,1,1)$ is closest to the point $b = (2,4,4)$? Find also the point closest to a on the line through b .

Step 1: Write the projected vector

Let the projected vector be

$$p = xa$$

Step 2: Use the projection formula to find x

$$x = \frac{a^T b}{a^T a}$$

$x=10/3$. $10/3$ times a is closest to point b .

Point on the line through b closest to a

Step 1: Find the projected vector of a on b

$$p = \frac{b^T a}{b^T b} b$$

projection of a on b is

$$p = \left(\frac{5}{9}, \frac{10}{9}, \frac{10}{9}\right)$$

is the point on line through b closest to a

10. What matrix P projects every point in $\mathbb{R}^3$ onto the line of intersection of the planes $x+y+t=0$ and $x-t=0$?

Step 1: Find the vector of line of intersection of $x+y+t=0$ and $x-t=0$

Vector of the Line is

$$v = (1, -2, 1)$$

Steps 2: Find the Projection matrix that project every point in $\mathbb{R}^3$ on to v

For projection onto a line spanned by v

$$P = \frac{vv^T}{v^T v}$$

$$P = \frac{1}{6} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

This matrix projects every point in $\mathbb{R}^3$ onto the line of intersection of the two planes.

Chapter 3.3

Find the least squares solution x of the system $Ax = b$ and determine the vector $p = Ax$, where

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Then verify that the error vector $e = b - p$ is orthogonal to the columns of A .

Solution

Step1: We solve the least squares problem $Ax = b$ using the normal equations:

$$\begin{aligned} A^T Ax &= A^T b \\ \text{Compute } x &= (A^T A)^{-1} A^T b \\ x &= \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix} \end{aligned}$$

Step2: Compute the projection $p = Ax$

$$p = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$$

Step 3: Compute the error $e = b - p$

$$e = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}$$

Step 4: Verify e is perpendicular to columns of A

Columns of A :

$$a_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, a_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Check if $a_1 \cdot e = 0$ and $a_2 \cdot e = 0$

Final Answer

$$x = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \\ 3 \end{bmatrix}, p = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 3 \\ 2 \\ 3 \end{bmatrix}$$

and the error $b - p$ is orthogonal to the columns of A .

2. Suppose the data values $b_1 = 1$ and $b_2 = 7$ are observed at times $t_1 = 1$ and $t_2 = 2$, respectively. A line of the form $b = Dt$ is used to fit the data, where the line passes through the origin.

Find the value of D that best fits the data using the least squares method, and sketch the resulting best-fit line.

Step 1: Write the system in matrix form

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, x = D, b = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

Step 2: Compute $x = (A^T A)^{-1} A^T b$

$$D=3$$

Best least squares line

$$b = 3t$$

Step 3: Sketch the best fit line

3. Find the projection of the vector b onto the column space of the matrix A , where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}.$$

Then decompose b into two components $b = p + q$, where p lies in the column space of A and q is orthogonal to the column space.

Finally, determine which of the four fundamental subspaces of A contains the vector q .

Step 1: Solve for $\hat{x}$

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$\hat{x} = \begin{bmatrix} \frac{9}{22} \\ \frac{37}{22} \end{bmatrix}$$

Step 2: Projection $p = A\hat{x}$

$$p = \begin{bmatrix} \frac{23}{11} \\ -\frac{14}{11} \\ \frac{65}{11} \end{bmatrix}$$

Step 3: Split $b = p + q$

$$p = \begin{bmatrix} \frac{23}{11} \\ \frac{14}{11} \\ -\frac{65}{11} \end{bmatrix}, q = \begin{bmatrix} \frac{12}{11} \\ -\frac{36}{11} \\ \frac{12}{11} \end{bmatrix}$$

p is in Column Space of A , the vector q is perpendicular to the column space.

Therefore

$$q \in N(A^T)$$

4. The vectors $a_1 = (1,1,0)$ and $a_2 = (1,1,1)$ span a plane in $\mathbb{R}^3$.

1. Find the projection matrix P that projects vectors in $\mathbb{R}^3$ onto this plane.
2. Find a nonzero vector b whose projection onto this plane is zero.

Step 1: The plane is spanned by

$$a_1 = (1,1,0), a_2 = (1,1,1)$$

So

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The projection matrix onto the column space of A is

$$P = A(A^T A)^{-1} A^T$$

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step2 : Find a nonzero vector projected to zero Such vectors lie in the orthogonal complement of the plane. Find the Left null space of the plane

$$b=(1,-1,0)$$

5. Find the projection matrix P onto the space spanned by $a_1=(1,0,1)$ and $a_2=(1,1,-1)$.

Step1: Find Projection matrix use the formula:

$$P = A(A^T A)^{-1} A^T$$

:

$$P = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} & \frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{6} & -\frac{1}{3} & \frac{5}{6} \end{bmatrix}$$

Find the weighted least-squares solution $\hat{x}_W$ to the system $Ax = b$, where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, W = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Use formula: $\hat{x}_W = (A^T W A)^{-1} A^T W b$

Solution:

$$\hat{x}_W = \begin{bmatrix} \frac{1}{11} \\ \frac{6}{11} \end{bmatrix}$$

Chapter 3.4 The Gram-Schmidt Process

1. Given the non-orthogonal vectors

$$a = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, c = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

find an orthonormal set of vectors q_1, q_2, q_3 obtained from a, b, c .

Step 1: Find q_1

$$q_1 = \frac{a}{\|a\|}$$

Step 2: Find q_2

$$u_2 = b - (q_1^T b)q_1$$

$$q_2 = \frac{u_2}{\|u_2\|}$$

Step 3: Find q_3

:

$$u_3 = c - (q_1^T c)q_1 - (q_2^T c)q_2$$

$$q_3 = \frac{u_3}{\|u_3\|}$$

Orthonormal Set

$$q_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, q_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ 1 \\ -\frac{1}{\sqrt{6}} \\ 2 \\ \frac{1}{\sqrt{6}} \end{bmatrix}, q_3 = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

2. Project the vector

$$b = (0, 3, 0)$$

onto each of the orthonormal vectors

$$a_1 = \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right), a_2 = \left(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right).$$

1. Find the projections of b onto a_1 and a_2 .
2. Find the projection of b onto the plane spanned by a_1 and a_2 .

Step 1: Project onto a_1

Step 2: Project onto a_2

Step 3: Projection onto the plane

$$p = \text{proj}_{a_1}(b) + \text{proj}_{a_2}(b)$$

Projection onto each vector:

$$\text{proj}_{a_1}(b) = \left(\frac{4}{3}, \frac{4}{3}, -\frac{2}{3}\right)$$

$$\text{proj}_{a_2}(b) = \left(-\frac{2}{3}, \frac{4}{3}, \frac{4}{3}\right)$$

Projection onto the plane:

$$p = \left(\frac{2}{3}, \frac{8}{3}, \frac{2}{3}\right)$$

3. Find a third column q_3 such that the matrix

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & ? \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & ? \\ \frac{1}{\sqrt{3}} & -\frac{3}{\sqrt{14}} & ? \end{bmatrix}$$

is an orthogonal matrix.

The third column must be:

- a unit vector, and
 - orthogonal to the first two columns.
1. Find such a vector q_3 .
 2. How much freedom do you have in choosing q_3 ?
 3. Verify that the rows of Q are automatically orthonormal.

Step1: Find Left Null space of matrix with column 1 and column2.

Step 2: divide basis vector of Left Null space by its magnitude. This Vector is orthonormal to two columns.

third column

$$q_3 = \begin{bmatrix} -\frac{5}{\sqrt{42}} \\ 4 \\ \frac{1}{\sqrt{42}} \end{bmatrix}$$

How much freedom?

Two choices: q_3 or $-q_3$

Because:

- Both are unit vectors
- Both are orthogonal to q_1, q_2

Why rows also become orthonormal?

If Q has orthonormal columns:

$$Q^T Q = I$$

For a square matrix, this implies:

$$Q Q^T = I$$

So rows are automatically orthonormal.

-
4. Find an orthonormal set of vectors q_1, q_2, q_3 such that q_1 and q_2 span the column space of

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}.$$

1. Identify which fundamental subspace contains q_3 .

2. Find the least-squares solution of $Ax = b$, where

$$b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}.$$

Step 1: Find

$$q_1 = \frac{a}{\|a\|}$$

Step 2: Find q_2

$$u_2 = b - (q_1^T b)q_1$$

$$q_2 = \frac{u_2}{\|u_2\|}$$

Step 3: $q_3 \in \text{Null}(A^T)$ Find basis vector $\text{Null}(A^T)$

$u_3 = \text{basis vector Null}(A^T)$

$$q_3 = \frac{u_3}{\|u_3\|}$$

Step 4: Least-squares solution

Use:

$$Rx = Q^T b$$

Final Results

$$\bullet \quad q_1 = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}$$

$$\bullet \quad q_2 = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$$

$$\bullet \quad q_3 = \begin{bmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

$q_3 \in \text{Left Null Space}$

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Express the Gram–Schmidt orthogonalization of the vectors a and b in the form $A = QR$, where

$$a = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

1. Find matrices Q and R .
2. For n vectors a_i each having m components, determine the shapes (dimensions) of A , Q , and R .

Step 1: Find

$$q_1 = \frac{a}{\|a\|}$$

Step 2: Find q_2

$$u_2 = b - (q_1^T b)q_1$$

$$q_2 = \frac{u_2}{\|u_2\|}$$

Step 3: Form Q

Step 4: Form R

$$R = Q^T A$$

Solution:

$$\boxed{A = QR}$$

$$Q = \begin{bmatrix} \frac{1}{3} & 0 \\ \frac{2}{3} & \frac{1}{\sqrt{2}} \\ \frac{2}{3} & -\frac{1}{\sqrt{2}} \end{bmatrix}, R = \begin{bmatrix} 3 & 3 \\ 0 & \sqrt{2} \end{bmatrix}$$

Use the Gram–Schmidt process to find orthogonal vectors A, B, C from the vectors

$$a = (1, -1, 0, 0), b = (0, 1, -1, 0), c = (0, 0, 1, -1).$$

Show that both sets $\{A, B, C\}$ and $\{a, b, c\}$ form bases for the subspace of vectors in $\mathbb{R}^4$ that are perpendicular to

$$d = (1,1,1,1).$$

Step 1: Find q_1

$$A = \frac{a}{\|a\|}$$

Step 2: Find q_2

$$B = b - (q_1^T b)q_1$$

$$q_2 = \frac{u_2}{\|u_2\|}$$

Step 3: Find q_3

:

$$C = c - (q_1^T c)q_1 - (q_2^T c)q_2$$

Step 4: Check if $A \cdot B=0, A \cdot C=0, B \cdot C=0, C \cdot D=0$

Check if $a \cdot d=0, b \cdot d=0, c \cdot d=0$

Orthogonal vectors

$$A = (1, -1, 0, 0)$$

$$B = \left(\frac{1}{2}, \frac{1}{2}, -1, 0\right)$$

$$C = \left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right)$$

7. Apply Gram-Schmidt orthogonalization process to find an orthogonal basis and

then an orthonormal basis for the subspace U of $\mathbb{R}^4$ spanned by a) $v_1=(1,1,1,1)$, b) $v_2=(1,2,4,5)$, c) $v_3=(1,-3,-4,-2)$

Step 1: Find q_1

$$q_1 = \frac{a}{\|a\|}$$

Step 2: Find q_2

$$u_2 = b - (q_1^T b)q_1$$

$$q_2 = \frac{u_2}{\|u_2\|}$$

Step 3: Find q_3

:

$$u_3 = c - (q_1^T c)q_1 - (q_2^T c)q_2$$

$$q_3 = \frac{u_3}{\|u_3\|}$$

Orthonormal Basis

$$q_1 = \frac{1}{2}(1,1,1,1), q_2 = \frac{1}{\sqrt{10}}(-2, -1,1,2), q_3 = \frac{1}{\sqrt{910}}(16, -17, -13,14)$$

Chapter 5.1 Eigen Vector and Eigen values

1. If we shift a matrix A to $B = A - 7I$, how do the eigenvalues and eigenvectors change?

Given

$$B = A - 7I = \begin{bmatrix} -6 & -1 \\ 2 & -3 \end{bmatrix},$$

find the eigenvalues and eigenvectors of B , and relate them to those of A .

If we shift a matrix A to $B = A - 7I$, then:

Given

$$B = \begin{bmatrix} -6 & -1 \\ 2 & -3 \end{bmatrix}$$

Find A :

$$A = \begin{bmatrix} -6+7 & -1 \\ 2 & -3+7 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$$

Step 1: Find Eigenvalues of A and B

$$\det(A - \lambda I) = 0$$

Eigen values of A : 3,2

Eigenvalues of B : $-4, -5$

Step 2: Eigenvectors

Find Eigenvectors for A, B

$$(A - \lambda_i I)x = 0$$

Eigenvectors of A :

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Eigenvectors of B :

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Eigenvectors are the same for both matrices.

Eigen values will increase or decrease based on the multiplier of I

Find the eigenvalues and corresponding eigenvectors of the following matrices:

$$A = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}.$$

Further, verify that:

1. The sum of the eigenvalues is equal to the trace of the matrix.
2. The product of the eigenvalues is equal to the determinant of the matrix.

Step 1: Find Eigenvalues

Compute:

$$\det(A - \lambda I) = 0$$

$$\lambda = 3, 1, 0$$

Trace:

$$3 + 1 + 0 = 4 = \text{trace}(A)$$

- Determinant:

$$3 \cdot 1 \cdot 0 = 0 = \det(A)$$

Repeat same steps for B matrix. Solution for B

- Eigenvalues: $2, 2, -2$
- Trace=2
- Determinant=-8

Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

and the matrix

$$A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}.$$

A+I has the _____ eigenvectors as A. Its eigenvalues are _____ by 1.

Same as problem 1.

Find the eigenvalues of the following matrices:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

A is uppertriangular.

Step1: Compute for B and C:

$$\det(A - \lambda I) = 0$$

Eigenvalues of A: 1, 4, 6

Eigenvalues of B: $2, \sqrt{3}, -\sqrt{3}$

Eigenvalues of C: 6, 0, 0

Diagonalize the following matrices by factoring them in the form $A = S\Lambda S^{-1}$,

$$A_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}.$$

Find Eigen values

Find the Eigen vectors

Form Diagonal Matrix Λ : Place eigenvalues on the diagonal in descending order.

Place eigenvectors as columns of S in the same order as eigen values.

For $A_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$S = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$S^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

For $A_2 = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$

$$S = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}, \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & -\frac{1}{2} \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & -\frac{1}{2} \end{bmatrix}$$

Determine which of the following matrices are **not diagonalizable**:

$$A_1 = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}, A_2 = \begin{bmatrix} 2 & 0 \\ 2 & -2 \end{bmatrix}, A_3 = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}.$$

Find eigen values.

If any eigen value=0 then Not diagonalizable, eigen values unique

2X2 matrix should result in 2 independent eigen vectors

Not diagonalizable: A_1, A_3

Diagonalizable: A_2

verify the **Cayley–Hamilton theorem** for

$$A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$$

Step 1: Characteristic polynomial

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 5 \\ 4 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda) - 20 = \lambda^2 - 3\lambda - 18$$

Step 2: Cayley–Hamilton equation

$$A^2 - 3A - 18I = 0$$

Step 3: Compute A^2

$$A^2 = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 21 & 15 \\ 12 & 24 \end{bmatrix}$$

Step 4: Compute $3A$

$$3A = \begin{bmatrix} 3 & 15 \\ 12 & 6 \end{bmatrix}$$

Step 5: Compute $A^2 - 3A$

$$= \begin{bmatrix} 21 & 15 \\ 12 & 24 \end{bmatrix} - \begin{bmatrix} 3 & 15 \\ 12 & 6 \end{bmatrix} = \begin{bmatrix} 18 & 0 \\ 0 & 18 \end{bmatrix}$$

Step 6: Subtract $18I$

$$A^2 - 3A - 18I = \begin{bmatrix} 18 & 0 \\ 0 & 18 \end{bmatrix} - \begin{bmatrix} 18 & 0 \\ 0 & 18 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Final result

$$A^2 - 3A - 18I = 0$$

Hence verified.

Verify Cayley Hamilton for

$$A = \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix}$$

Step 1: Characteristic polynomial

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & -2 \\ -2 & -1-\lambda \end{vmatrix} \\ &= (2-\lambda)(-1-\lambda) - 4 = \lambda^2 - \lambda - 6 \end{aligned}$$

Step 2: Cayley–Hamilton equation

$$A^2 - A - 6I = 0$$

Step 3: Compute A^2

$$A^2 = \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix}$$

Step 4: Compute $A^2 - A$

$$= \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix} - \begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

Step 5: Subtract $6I$

$$A^2 - A - 6I = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} - \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Final result

$$A^2 - A - 6I = 0$$

Hence verified.

Find inverse of A using Cayley Hamilton

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 9 & 2 & 0 \\ 5 & 0 & 3 \end{bmatrix}$$

Step 1: Characteristic polynomial

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 & 2 \\ 9 & 2-\lambda & 0 \\ 5 & 0 & 3-\lambda \end{vmatrix}$$

$$\begin{aligned}
 &= (1 - \lambda)(2 - \lambda)(3 - \lambda) - 9(3 - \lambda) - 10(2 - \lambda) \\
 &= -\lambda^3 + 6\lambda^2 + 8\lambda - 41
 \end{aligned}$$

So,

$$\lambda^3 - 6\lambda^2 - 8\lambda + 41 = 0$$

Step 2: Cayley–Hamilton equation

$$A^3 - 6A^2 - 8A + 41I = 0$$

Step 3: Multiply by A^{-1}

$$\begin{aligned}
 A^2 - 6A - 8I + 41A^{-1} &= 0 \\
 41A^{-1} &= -A^2 + 6A + 8I \\
 A^{-1} &= \frac{1}{41}(-A^2 + 6A + 8I)
 \end{aligned}$$

Step 4: Compute A^2

$$A^2 = \begin{bmatrix} 20 & 3 & 8 \\ 27 & 13 & 18 \\ 20 & 5 & 19 \end{bmatrix}$$

Step 5: Compute expression

$$\begin{aligned}
 -A^2 &= \begin{bmatrix} -20 & -3 & -8 \\ -27 & -13 & -18 \\ -20 & -5 & -19 \end{bmatrix} \\
 6A &= \begin{bmatrix} 6 & 6 & 12 \\ 54 & 12 & 0 \\ 30 & 0 & 18 \end{bmatrix} \\
 8I &= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}
 \end{aligned}$$

Add:

$$-A^2 + 6A + 8I = \begin{bmatrix} -6 & 3 & 4 \\ 27 & 7 & -18 \\ 10 & -5 & 7 \end{bmatrix}$$

Final answer

$$A^{-1} = \frac{1}{41} \begin{bmatrix} -6 & 3 & 4 \\ 27 & 7 & -18 \\ 10 & -5 & 7 \end{bmatrix}$$



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UE24MA241B - Linear Algebra and its Applications

Unit 4 Classwork problem and Solution

Q1

(a) What 3 by 3 symmetric matrices A_1 and A_2 correspond to f_1 and f_2 ?

$$f_1 = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$

$$f_2 = x_1^2 + 2x_2^2 + 11x_3^2 - 2x_1x_2 - 2x_1x_3 - 4x_2x_3.$$

For a quadratic form

$f(x) = x^T Ax$, the symmetric matrix A is obtained by:

- Diagonal entries = coefficients of x_i^2
- Off-diagonal entries $a_{ij} = \frac{\text{coefficient of } x_i x_j}{2}$

○

$$A_1 = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & -2 \\ -1 & -2 & 11 \end{bmatrix}$$

Q2. Check for positive definiteness of the following

(a) $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$, (b) $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, (c) $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, (d) $\begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix}$.

Use anyone method pivot, eigen value, determinant of upper left submatrix, $R^T R$, $X^T A X$.

a,b,d are not positive definite

c is positive definite

8. If $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite, test $A^{-1} = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$ for positive definiteness.

If A eigen values are e_1, e_2 , A^{-1} eigen values will be $1/e_1, 1/e_2$

e_1 and e_2 will be positive and non zero therefore $1/e_1, 1/e_2$ will also be positive and nonzero

2. Decide for or against the positive definiteness of

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}^2$$

A through pivot method, B through determinant method and C through eigen values.

Matrix A (Pivot method)

Apply Gaussian elimination (no row swaps):

- Pivot 1: $2 > 0$
- Pivot 2: $\frac{3}{2} > 0$
- Pivot 3: 0

Since last pivot is **zero**, all pivots are not positive

A is NOT positive definite

Matrix B (Determinant method)

Check leading principal minors/determinant of upper left submatrices:

$$D_1 = 2 > 0$$

$$D_2 = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 4 - 1 = 3 > 0$$

$$D_3 = \det(B) = 4 > 0$$

B is Positive Definite

Matrix C^2 (Eigenvalue method)

$$\det(C - \lambda I) = 0$$

Solving gives eigenvalues:

$$\lambda_1 = 3, \lambda_2 = -1, \lambda_3 = -2$$

Eigen value of C^2

$$\lambda_1 = 9, \lambda_2 = 1, \lambda_3 = 4$$

Since eigenvalues are positive

C^2 is positive definite

1. Find the Singular Value Decomposition (SVD) of the matrix

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

$$A = U\Sigma V^T$$

Step 1: Compute AA^T

$$AA^T = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

Step 2: Eigenvalues of AA^T

$$\det \begin{bmatrix} 17 - \lambda & 8 \\ 8 & 17 - \lambda \end{bmatrix} = (17 - \lambda)^2 - 64 = 0$$

$$\Rightarrow \lambda = 25, 9$$

Singular values:

$$\sigma_1 = 5, \sigma_2 = 3$$

Step 3: Find U (eigenvectors of AA^T)

For $\lambda = 25$:

$$\begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \Rightarrow u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda = 9$:

$$\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \Rightarrow u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Step 4: Find V using $v_i = \frac{1}{\sigma_i} A^T u_i$

For $\sigma_1 = 5$

$$v_1 = \frac{1}{5} A^T u_1 = \frac{1}{5\sqrt{2}} \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

For $\sigma_2 = 3$

$$v_2 = \frac{1}{3} A^T u_2 = \frac{1}{3\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ -4 \end{bmatrix}$$

Find v_3 using Orthogonality conditions

$$v_3 \cdot v_1 = 0 \Rightarrow \frac{1}{\sqrt{2}}(x + y) = 0 \Rightarrow x + y = 0 \Rightarrow y = -x$$

$$v_3 \cdot v_2 = 0 \Rightarrow \frac{1}{3\sqrt{2}}(x - y + 4z) = 0 \Rightarrow x - y + 4z = 0$$

Solving the equations

$$v_3 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$$

Normalize:

$$v_3 = \frac{1}{3} \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

Final SVD

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$
$$\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}$$
$$V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{4}{3\sqrt{2}} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

1. Find the Singular Value Decomposition (SVD) of the matrix

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}.$$

Step 1: Compute $A^T A$

$$A^T A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 9 & -1 \\ -1 & 9 \end{bmatrix}$$

Step 2: Eigenvalues of $A^T A$

$$\det \begin{bmatrix} 9 - \lambda & -1 \\ -1 & 9 - \lambda \end{bmatrix} = (9 - \lambda)^2 - 1 = 0$$
$$\Rightarrow \lambda = 10, 8$$

Step 3: Find V (eigenvectors of $A^T A$)

For $\lambda = 10$:

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \Rightarrow v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

For $\lambda = 8$:

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

Step 4: Find U using $u_i = \frac{1}{\sigma_i} A v_i$

For $\sigma_1 = \sqrt{10}$

$$A v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix}$$
$$u_1 = \frac{1}{\sqrt{10}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{5}} \\ 2 \\ \frac{\sqrt{5}}{0} \end{bmatrix}$$

For $\sigma_2 = 2\sqrt{2}$

$$Av_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

$$u_2 = \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Find u_3

$$u_3 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}$$

Final SVD

$$U = \begin{bmatrix} -\frac{1}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} \\ 2 & 0 & 1 \\ \frac{1}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 2\sqrt{2} \\ 0 & 0 \end{bmatrix}$$

$$V^T = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

Step 1: Check Positive Definiteness

Use **leading principal minors**:

- $D_1 = 3 > 0$
- $D_2 = \det(A) = 3 \cdot 2 - (-1)(-1) = 6 - 1 = 5 > 0$

Eigenvalues

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 3-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = (3-\lambda)(2-\lambda) - 1 = 0$$

$$\lambda^2 - 5\lambda + 5 = 0$$

$$\lambda_1 = \frac{5 + \sqrt{5}}{2}, \lambda_2 = \frac{5 - \sqrt{5}}{2}$$

Eigenvectors

For $\lambda_1 = \frac{5 + \sqrt{5}}{2}$

From $(A - \lambda I)x = 0$:

$$x_2 = \frac{1 - \sqrt{5}}{2} x_1$$

Take $x_1 = 1$:

$$v_1 = \begin{bmatrix} 1 \\ \frac{1 - \sqrt{5}}{2} \end{bmatrix}$$

For $\lambda_2 = \frac{5 - \sqrt{5}}{2}$

$$x_2 = \frac{1 + \sqrt{5}}{2} x_1$$

$$v_2 = \begin{bmatrix} 1 \\ \frac{1 + \sqrt{5}}{2} \end{bmatrix}$$

Normalize Eigenvectors

After normalization:

$$Q = \begin{bmatrix} \frac{1}{\sqrt{1 + \left(\frac{1 - \sqrt{5}}{2}\right)^2}} & \frac{1}{\sqrt{1 + \left(\frac{1 + \sqrt{5}}{2}\right)^2}} \\ \frac{\frac{1 - \sqrt{5}}{2}}{\sqrt{1 + \left(\frac{1 - \sqrt{5}}{2}\right)^2}} & \frac{\frac{1 + \sqrt{5}}{2}}{\sqrt{1 + \left(\frac{1 + \sqrt{5}}{2}\right)^2}} \end{bmatrix}$$

Final Answer

$$A = Q\Lambda Q^T$$

where

$$\Lambda = \begin{bmatrix} \frac{5 + \sqrt{5}}{2} & 0 \\ 0 & \frac{5 - \sqrt{5}}{2} \end{bmatrix}$$

Given the data matrix

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},$$

Row as feature

perform Principal Component Analysis (PCA) using Singular Value Decomposition (SVD)

Step 1: Center the matrix X_c

Step 2: Check if X_c is positive definite. It is not positive definite

Step3: Find SVD using $X_c X_c^T$

$$U = V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

Given the dataset

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 3 & 1 \end{bmatrix},$$

Column as feature

perform Principal Component Analysis (PCA) using the covariance matrix method

Step 1: Compute Mean of Each Column

Step 2: Center the Data

Step 3: Covariance Matrix

Step 4: Find Eigenvalues and eigenvectors

Principal Components

$$PC_1 = \begin{bmatrix} 0.882 \\ -0.472 \end{bmatrix}$$

$$PC_2 = \begin{bmatrix} 0.472 \\ 0.882 \end{bmatrix}$$

From the pivots, eigenvalues, and eigenvectors of $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$, write A as $R^T R$ in three ways: $(L\sqrt{D})(\sqrt{D}L^T)$, $(Q\sqrt{\Lambda})(\sqrt{\Lambda}Q^T)$, and $(Q\sqrt{\Lambda}Q^T)(Q\sqrt{\Lambda}Q^T)$.

Find L D U

$$A = L\sqrt{D}\sqrt{D}L^T = (\sqrt{D}L^T)^T(\sqrt{D}L^T) = R^T R$$

$$\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4/5 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 \\ 0 & 3/\sqrt{5} \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 \\ 0 & 3/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1 & 4/5 \\ 0 & 1 \end{bmatrix}$$

Find Normalized Eigen vectors and Q

$$A = Q\sqrt{\Lambda}\sqrt{\Lambda}Q^T$$

with

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \sqrt{\Lambda} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Spectral Form

$$A = Q\Lambda Q^T$$

with

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \Lambda = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$$

Square Root Form

$$A^{1/2} = Q\Lambda^{1/2}Q^T$$

$$A = (Q\Lambda^{1/2}Q^T)^2$$

